

DIPOLE - DIPOLE INTERACTION - SI UNITS

$$\phi(r) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\vec{r} - \vec{r}_1|} - \frac{1}{|\vec{r}_1|} \right) \quad (1)$$

Expanding for $|\vec{r}_1|$ small

$$\frac{1}{|\vec{r} - \vec{r}_1|} = \frac{1}{|\vec{r}|} + \frac{\vec{r} \cdot \vec{r}_1}{[r^2]^{3/2}} + O(r_1^2) \quad (2)$$

$$\phi(r) = \frac{q}{4\pi\epsilon_0} \frac{\vec{r} \cdot \vec{r}_1}{r^3} = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3} \quad (3)$$

$$\vec{E} = -\nabla \phi(r) = -\nabla \left[\frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \vec{r}}{r^3} \right] \quad (4)$$

$$E_i = -\frac{\partial}{\partial r_i} \frac{\vec{p} \cdot \vec{r}}{4\pi\epsilon_0 r^3} = -\frac{\partial}{\partial r_i} \frac{\sum_j p_j r_j}{4\pi\epsilon_0 r^3} \quad (5)$$

$$= -\frac{p_j}{4\pi\epsilon_0} \left[\frac{1}{r^3} \frac{\partial r_j}{\partial r_i} - r_j \frac{\partial}{\partial r_i} \frac{1}{r^3} \right]$$

$$E_i = T_{ij} p_j \quad (6)$$

	DISCRETE FORMULATION
$T_{ij} = \begin{cases} \frac{1}{r^3} \left[\frac{3r_i r_j}{r^2} - \delta_{ij} \right] \frac{1}{4\pi\epsilon_0} & r > 0 \\ \frac{1}{3\epsilon_0} \delta(r) \delta_{ij} \times \frac{1}{4\pi\epsilon_0} & r = 0 \end{cases}$	(7)

In the continuum notation

$$\vec{\nabla} \cdot (\vec{r}_a - \vec{r}_b) = \frac{1}{4\pi\epsilon_0} \left\{ \frac{3(\vec{r}_a - \vec{r}_b) \cdot (\vec{r}_a - \vec{r}_b)}{r^5} - \frac{I}{r^3} \right\}$$

$$\boxed{\vec{r}_a - \vec{r}_b > 0}$$

$$\vec{\nabla} \cdot (\vec{r}_a - \vec{r}_b) = \frac{1}{4\pi\epsilon_0} \delta(r) I \quad \vec{r}_a - \vec{r}_b = 0$$

CONTINUUM

DIELECTRIC CONSTANT CALCULATIONS

$\vec{E}_0$

In the presence of an external field and dielectric medium

(9)

$$\vec{E} = \vec{E}_0 + \int \frac{\vec{r} \cdot \vec{P}(r')}{|\vec{r} - \vec{r}'|^3} d^3r'$$

$$\vec{E} = \vec{E}_0 \quad r=0 \quad (10)$$

$$\vec{E}(\vec{r} - \vec{r}') = \frac{1}{4\pi\epsilon_0} \left(\frac{3(\vec{r} - \vec{r}') \cdot (\vec{r} - \vec{r}')}{r^5} - \frac{I}{r^3} \right)$$

$$\varepsilon = (1 + \bar{x})$$

$$(IV) \quad \bar{x} = \left(\frac{\varepsilon - 1}{\varepsilon} \right)$$

$$\vec{P} = \bar{x} \varepsilon_0 \vec{E}$$

$$= x \varepsilon_0 \left\{ \vec{E}_0 + \int \vec{T}(\vec{r} - \vec{r}') \cdot \vec{P}(\vec{r}') d^3 r' \right\}$$

$$= x \varepsilon_0 \left\{ \vec{E}_0 + \vec{E}_1 + \vec{E}_2 \right\} \quad (12)$$

$$\vec{E}_1 = \int_{|\vec{r} - \vec{r}'| < R} \vec{T}(\vec{r} - \vec{r}') \cdot \vec{P}(\vec{r}') d^3 r' \quad (13)$$

$$\vec{E}_2 = \int_{|\vec{r} - \vec{r}'| > R} \vec{T}(\vec{r} - \vec{r}') \cdot \vec{P}(\vec{r}') d^3 r' \quad (14)$$

$$\begin{aligned} \lim_{R \rightarrow 0} \int_{|\vec{r} - \vec{r}'| < R} \frac{1}{4\pi \varepsilon_0} \delta(\vec{r} - \vec{r}') \vec{I} \cdot \vec{P}(\vec{r}') d^3 r' &= \vec{E}_1 \\ &= \frac{1}{4\pi \varepsilon_0} \int_{|\vec{r} - \vec{r}'| < R} \delta(\vec{r} - \vec{r}') \vec{I} \cdot \vec{P}(\vec{r}') d^3 r' \quad (15) \end{aligned}$$

$$\vec{E}_1 = -\frac{1}{3\varepsilon_0} \vec{P}(\vec{r}) \quad (16)$$

$$\vec{P}(\vec{r}) = \chi \epsilon_0 \vec{E}_0 - \frac{1}{3\epsilon_0} \chi \epsilon_0 \vec{P}(\vec{r}) + \chi \epsilon_0 \vec{E}_2 \quad (17)$$

$$\vec{P} \left[1 + \frac{\chi}{3} \right] = \chi \epsilon_0 \vec{E}_0 + \chi \epsilon_0 \vec{E}_2$$

$$\vec{P} \left[1 + \frac{(\epsilon - 1)}{3} \right] = \chi \epsilon_0 [\vec{E}_0 + \vec{E}_2]$$

$$\vec{P} \left[\frac{3 + \epsilon - 1}{3} \right] = \chi \epsilon_0 [\vec{E}_0 + \vec{E}_2]$$

$$\vec{P} \left[\frac{2 + \epsilon}{3} \right] = \chi \epsilon_0 [\vec{E}_0 + \vec{E}_2] \quad (18)$$

$$\vec{P}(\vec{r}) = 3\epsilon_0 \frac{\epsilon - 1}{\epsilon + 2} [\vec{E}_0(\vec{r}) + \vec{E}_2(\vec{r})] \quad (19)$$

General solution

$$\vec{P}(\vec{r}) = 3\varepsilon_0 \frac{\varepsilon-1}{\varepsilon+2} \left[\vec{E}_0(\vec{r}) + \vec{E}_2(\vec{r}) \right] \quad (20)$$

$$\vec{E}_2(\vec{r}) = \int_{|\vec{r}-\vec{r}'| > R} \frac{\vec{T}(\vec{r}-\vec{r}')}{|\vec{r}-\vec{r}'|^3} \cdot \vec{P}(\vec{r}') d^3r'$$

$$\vec{E}_2(\vec{k}) = \vec{T}(\vec{k}) \cdot \vec{P}(\vec{k}) \quad (21)$$

Using (21) in (20)

$$\vec{P}(\vec{k}) = 3\varepsilon_0 \frac{\varepsilon-1}{\varepsilon+2} \left[\vec{E}_0(\vec{k}) + \vec{T}(\vec{k}) \cdot \vec{P}(\vec{k}) \right]$$

$$\vec{P}(\vec{k}) \cdot \left[\vec{I} - 3\varepsilon_0 \frac{\varepsilon-1}{\varepsilon+2} \vec{T}(\vec{k}) \right] = 3\varepsilon_0 \frac{\varepsilon-1}{\varepsilon+2} \vec{E}_0(\vec{k})$$

$\vec{I} \rightarrow \parallel \vec{P}$
 $\vec{I} \rightarrow \text{UNIT VECTOR}$

$$\vec{P}(\vec{k}) = 3\varepsilon_0 \frac{\varepsilon-1}{\varepsilon+2} \frac{\vec{E}_0(\vec{k})}{\left[\vec{I} - 3\varepsilon_0 \frac{\varepsilon-1}{\varepsilon+2} \vec{T}(\vec{k}) \right]}$$

(22)

THE DENOMINATOR HAS A DECIMAL
PROBLEM. THIS CAN BE FIXED WHEN

CALCULATING.

$$\langle \vec{M} \rangle_{\vec{E}_0} = \frac{\int d^3 r d^3 p e^{-\beta(\vec{H}_0 - \vec{E}_0 \cdot \vec{M})}}{\int d^3 r d^3 p e^{-\beta(\vec{H}_0 - \vec{E}_0 \cdot \vec{M})}}$$

The x component for small $\vec{E}_0$ (†)

$$\begin{aligned} \langle \vec{M} \rangle_{\vec{E}_0}^x &= \frac{\int d^3 r d^3 p e^{-\beta \vec{H}_0} M_x [1 + \beta \vec{E}_0 \cdot \vec{M}_x]}{\int d^3 r d^3 p e^{-\beta \vec{H}_0} [1 + \beta \vec{E}_0 \cdot \vec{M}_x]} + O(E_0^2) \\ &= \frac{\int d^3 r d^3 p e^{-\beta \vec{H}_0} M_x}{\int d^3 r d^3 p e^{-\beta \vec{H}_0}} + \frac{\int d^3 r d^3 p \beta \vec{E}_0 \cdot \vec{M}_x^2}{\int d^3 r d^3 p e^{-\beta \vec{H}_0}} \\ &= \beta \vec{E}_0^x \frac{\int d^3 r d^3 p e^{-\beta \vec{H}_0} M_x^2}{[\int d^3 r d^3 p e^{-\beta \vec{H}_0}]^2} \quad (23) \end{aligned}$$

Regarding the $\vec{E}_0$ should be independent of $\vec{r}$.

$$\langle \vec{M} \rangle_{\vec{E}_0}^x - \langle \vec{M} \rangle_0^x = \beta \vec{E}_0^x \langle M_x^2 \rangle - \beta \vec{E}_0^x \langle M_x \rangle^2 \quad (24)$$

$$\begin{aligned} \langle M_x^2 \rangle &= \frac{\langle M^2 \rangle}{3} \dots \\ \langle \Delta \vec{M} \rangle &= \beta \vec{E}_0 [\langle M^2 \rangle - \langle M \rangle^2] / 3 \quad (27) \end{aligned}$$

(†) Note that we do not consider

cross terms $M_x [1 + \beta \vec{E}_0 \cdot \vec{M}_x + \beta \vec{E}_0^x M_y + \beta \vec{E}_0^y M_x + \beta \vec{E}_0^z M_z]$

Furthermore,

$$\vec{P}(r) = \frac{\langle \Delta \vec{M}(r) \rangle}{V} \quad (28)$$

$$V \vec{P}(r) = \frac{\langle \Delta \vec{M}(r) \rangle}{3} = \beta \vec{E}_0(r) [\langle M_0^2 \rangle - \langle M^2 \rangle] \quad (29)$$

$$\vec{P}(\vec{r}) = \frac{\beta \vec{E}_0(r)}{3V} [\langle M(r)^2 \rangle - \langle M(r) \rangle^2] \quad (30)$$

$$\vec{P}(k=0) = \int \vec{P}(\vec{r}) d^3\vec{r}$$

$$\vec{P}(k=0) = \frac{\beta \vec{E}_0(k=0)}{3V} [\langle M^2 \rangle - \langle M \rangle^2] \quad (31)$$

Now comparing (31) with (22)

$$3\epsilon_0 \frac{\epsilon-1}{\epsilon-2} \frac{\vec{E}_0(k=0)}{[I - 3\epsilon_0 \frac{\epsilon-1}{\epsilon+2} T(k=0)]} = \frac{\beta \vec{E}_0(k=0) [\langle M^2 \rangle - \langle M \rangle^2]}{3V} \quad (32)$$

Simplifying

$$3\epsilon_0 \frac{\epsilon-1}{\epsilon-2} = \frac{\rho [\langle M^2 \rangle - \langle M \rangle^2]}{3V} [I - 3\epsilon_0 \frac{\epsilon-1}{\epsilon+2}]$$

SPHERICAL CUTOFF

$$\vec{E}_T(k=0) = \int d^3r \frac{\vec{r}}{r^3} \rho(r)$$

= 0 in spherical

$$3 \frac{\epsilon-1}{\epsilon-2} = \frac{[\langle M^2 \rangle - \langle M \rangle^2]}{3V k_B T \epsilon_0}$$

EWALD SUM [Preciso ainda por o arko go do Neumann]

$$3 \left(\frac{\epsilon-1}{\epsilon+2} \right) = \frac{[\langle M^2 \rangle - \langle M \rangle^2]}{3V k_B T \epsilon_0} \left(1 - \frac{3\epsilon_0 \epsilon-1}{\epsilon+2} \vec{F}(k=0) \right)$$

... (por fazer)

$$3\epsilon_0 \int d^3r \vec{F}(r) = Q = \int_0^{r_c} 4\pi r^2 \left(\frac{\rho}{r^{1/2}} \right) e^{-\eta^2 r^2} dr$$

$$\epsilon = 1 + \frac{[\langle M^2 \rangle - \langle M \rangle^2]}{3\epsilon_0 k_B T V}$$

Apparently $Q=1 \rightarrow$

